

PULLANI MADHU

[linkedin.com/in/madhu001](https://www.linkedin.com/in/madhu001) • madhupullani111@gmail.com • +91-9491163517 • Hyderabad, India (Open to Remote)

PROFESSIONAL SUMMARY

Generative AI / LLM Engineer with 2+ years at Tata Consultancy Services building production-grade AI systems. Delivered multi-agent LLM orchestration that reduced documentation effort by **50–60%** and code review cycles by **40%** across enterprise SAP deployments. Hands-on expertise in RAG pipeline design, LLaMA/Mistral fine-tuning with LoRA, LangGraph agent orchestration, hybrid vector search (FAISS + SAP HANA Vector), embeddings (OpenAI / HuggingFace), chain-of-thought and few-shot prompting, and enterprise integrations with SAP S/4HANA, ServiceNow, and Brainware. Targeting mid-level Generative AI / LLM Engineer roles in India or remote.

EXPERIENCE

Generative AI / LLM Engineer — **Tata Consultancy Services (TCS)** — *June 2024 – Present · Hyderabad, India*

- Designed and owned the end-to-end Functional Spec Agent pipeline on the iR2D platform — including RAG architecture with hybrid vector search (FAISS + SAP HANA Vector Engine), semantic re-ranking, OpenAI/HuggingFace embeddings, and LangGraph-based multi-agent orchestration for automated SAP document generation, contributing to a **50–60% reduction in documentation effort**.
- Reduced code development and review cycles by **40%** across the SDLC through agentic LLM workflows supporting GPT-4, Claude, and Gemini backends interchangeably.
- Built the **P2P Assistant** — a conversational AI chatbot enabling vendors to autonomously check PO status and raise ServiceNow tickets, integrating SAP S/4HANA REST APIs, Brainware document processing, and OAuth2-secured ERP data retrieval via FastAPI middleware.
- Developed the **Data Insights Assistant**, an NLP-to-SQL agent achieving **90% query translation accuracy** using chain-of-thought prompting with self-correction retry loops; delivered dynamic chart visualizations in a ReactJS dashboard reducing ad-hoc reporting from hours to seconds.
- Applied fine-tuning (LoRA) on LLaMA/Mistral models for instruction-following and domain-specific chat use cases; integrated NeMo Guardrails for output safety in production deployments.
- Engineered chain-of-thought, few-shot, and structured JSON output prompting strategies across multi-agent workflows to improve LLM output consistency and reduce hallucinations.

KEY PROJECTS

iR2D — Multi-LLM Orchestration Platform *2024 – Present*

Tech: LangGraph, LangChain, RAG, FAISS, SAP HANA Vector, OpenAI, HuggingFace Embeddings, NeMo Guardrails, SAP S/4HANA

- Architected a multi-LLM platform orchestrating GPT-4, Claude, and Gemini across user story processing, functional specs, code generation, code review, and test management — delivering **50–60%** documentation effort reduction and **30%** reduction in training content generation.
- Built hybrid RAG pipeline with chunking, indexing, OpenAI/HuggingFace embeddings, FAISS vector search, and semantic re-ranking; integrated SAP HANA Vector Engine for enterprise-scale retrieval augmentation.
- Deployed multi-cloud compatible architecture with ERP (SAP S/4HANA) connectivity and NeMo Guardrails for LLM output safety in production.

P2P Assistant — Vendor Self-Service Chatbot *Feb 2025 – Present*

Tech: LangChain, FastAPI, SAP S/4HANA REST APIs, ServiceNow, Brainware, OAuth2, Python

- Engineered a conversational AI assistant with enterprise-grade role-based access control and OAuth2 API authentication, eliminating manual procurement back-and-forth for vendors checking PO status or raising support tickets.
- Integrated SAP S/4HANA REST APIs, Brainware document processing, and ServiceNow ticketing workflows using FastAPI as the middleware layer; ensured compliance with data governance policies.

Data Insights — Intelligent Analytics Platform *Oct 2024 – Jan 2025*

Tech: LangGraph, SQL Agents, NLP, Chain-of-Thought Prompting, ReactJS, Python, PostgreSQL

- Built an NLP-to-SQL agentic platform with chain-of-thought prompting and self-correction retry loops, achieving **90% query-to-**

SQL translation accuracy for non-technical stakeholders.

- Delivered end-to-end pipeline: data ingestion → NLP query → SQL execution → automated narrative insights → dynamic ReactJS chart visualizations, reducing reporting turnaround from hours to seconds.

TECHNICAL SKILLS

Generative AI & LLMs: LangChain, LangGraph, CrewAI, OpenAI GPT-4, Claude, Gemini, LLaMA, Mistral, Ollama, HuggingFace Transformers

AI Techniques: RAG, Fine-tuning (LoRA), Embeddings, Multi-Agent Systems, Agentic Workflows, Chain-of-Thought Prompting, Few-Shot Prompting, Structured JSON Output, NeMo Guardrails, Cache Augmented Generation

Vector Databases: FAISS, SAP HANA Vector Engine, Chroma, Redis — hybrid vector search and semantic re-ranking

Machine Learning: Scikit-learn, TensorFlow, PyTorch, Keras, NLP, Computer Vision, Neural Networks

Cloud Platforms: Microsoft Azure (Azure OpenAI Service, Azure Machine Learning), AWS, Google Cloud Platform

Backend & APIs: Python, FastAPI, Flask, Django, Node.js, RESTful APIs, OAuth2, SAP S/4HANA REST APIs, ServiceNow

Frontend: ReactJS, HTML, CSS, JavaScript — dynamic chart visualizations (bar, pie, line)

Databases: PostgreSQL, MySQL, MongoDB, Neo4j

Data & Analytics: Pandas, NumPy, Matplotlib, Seaborn, Apache Spark, PowerBI, SQL

Languages: Python, JavaScript, SQL,

EDUCATION

B.Tech in Computer Science & Engineering — Vardhaman College of Engineering, Hyderabad — *Jul 2019 – Jun 2023*

CERTIFICATIONS

- Microsoft Certified: Azure AI Engineer Associate (AI-102) — Designing and Implementing Azure AI Solutions — 2025
- SAP Certified: Generative AI Developer — 2026